

Fig. 16: Dither frequency stabilisation

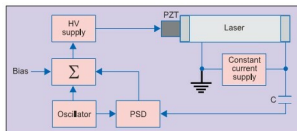


Fig. 17: Optogalvanic stabilisation schematic

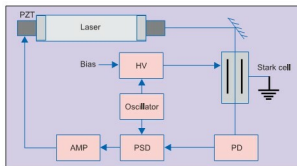


Fig. 18: Stark cell stabilisation block diagram

to the synchronous detector. The other input to the synchronous detector is the reference signal used to modulate the laser cavity length with the help of PZT. The error signal produced at the output of synchronous detector is used to stabilise the laser frequency.

Stark cell stabilisation technique allows dither-free frequency stabilisation. It also allows stabilising frequency anywhere on the Doppler broadened gain versus frequency curve of the lasers. Fig. 18 shows the block diagram of stark cell stabilisation.

A part of the output laser beam is made to pass through the stark cell. An appropriate value of DC voltage is applied to one of the stark plates to make the stark-shifted molecular transition of the gas contained in the cell precisely coincide with the carbon dioxide laser output line of interest. A slowly varying linear voltage

ramp may be applied instead. Also superimposed on this is a relatively high-frequency sinusoidal or square waveform. The frequency is 5kHz, typically. The beam at the output of the stark cell is thus modulated in both amplitude and frequency by the applied high-frequency signal around the line of interest.

The laser beam is phase-sensitively detected with high-frequency signal applied to the stark cell as reference. The output of phase-sensitive detector is the frequency-discriminant error signal. The error signal is amplified and then fed back to the PZT through a suitable interface. One of the end elements of the cavity is mounted on the PZT. The error signal adjusts the length of the cavity in the feedback mode to lock the laser frequency to the centre of the Doppler profile of the line of interest.

With this article, the four-part series focussing on the role of electron-

ics in military lasers and optoelectronics is completed. While introduction to role of electronics, solid state laser electronics and semiconductor diode laser electronics were discussed in the earlier parts, important building blocks of electronics used in common gas lasers are discussed in this concluding part. Different circuit topologies in use for design of power supplies for carbon dioxide and helium neon lasers have been discussed at length, highlighting important design-related issues with reference to gas lasers for military applications. Different techniques of frequency stabilisation of gas lasers, which is of paramount importance in applications such as laser radar, have also been covered. ●

*Concluded*

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